

Marawan Eldeib

M.Sc. Information Technology — University of Stuttgart

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SUMMARY

Software engineer and M.Sc. Information Technology student focused on application security and secure system design, with software-engineering experience at Fraunhofer IOSB. Skilled in web application penetration testing and building and hardening full-stack systems. Seeking a Master's thesis in security or software engineering from October 2026.

EDUCATION

M.Sc. Information Technology (Specialisation: Software Engineering) Oct 2024 – Present
University of Stuttgart Stuttgart, Germany

Relevant Coursework:

- **System & Web Application Security:** penetration testing, binary exploitation (buffer overflows, ROP), exploit mitigation (ASLR, DEP, stack canaries), and web security covering the OWASP Top 10 (XSS, SQL injection, CSRF, CORS, OAuth 2.0).
- **Distributed Systems:** engineered a fault-tolerant peer-to-peer messaging system with leader election, failure detection, and causal message ordering.
- **Cloud Computing:** architected a serverless, cost-optimised AWS platform (Lambda, DynamoDB, API Gateway, SQS, EventBridge) applying cloud security and the shared-responsibility model.

B.Eng. (Hons.) Electronics Majoring in Computer Jul 2020 – Oct 2024
Multimedia University Cyberjaya, Malaysia

CGPA: 3.7/4.0 — First Class Honours (German equivalent $\approx$ 1.4).

WORK EXPERIENCE

Working Student – Software Engineer Apr 2026 – Present
University of Stuttgart, Institute of Space Systems (IRS), SOFIA Data Center Stuttgart, Germany

- **Performance optimisation:** took a stalling NASA/DLR data-reduction job from over 50 GB (could not complete) to 14.4 GB with bit-identical output.
- **Root-cause analysis:** traced the peak to redundant concurrent calculations; a two-line config change gave 39%, and memray-guided dtype fixes a further 32%.

Working Student – Software Engineer Apr 2025 – Dec 2025
Fraunhofer IOSB Karlsruhe, Germany

- **Backend engineering:** built Java and Spring Boot backend services for LLM-based data extraction across Asset Administration Shell (AAS) digital-twin applications, benchmarking six large language models.
- **Test automation:** developed automated end-to-end test suites and curated 30+ evaluation datasets to validate model performance.

Intern – Software Engineer Jul 2023 – Oct 2023
AirAsia Kuala Lumpur, Malaysia

- **Feature delivery:** resolved 14 defects and shipped 5 checkout-calculation features, improving application stability for 10,000+ users.
- **Refactoring & APIs:** refactored 2 application modules and integrated GraphQL APIs for bilingual content management within Agile sprints.

PROJECTS

PrivacyLens — LLM Data-Exposure Analysis Apr 2026 – Present

Python | mitmproxy

- **Network traffic analysis:** measuring real-world data exposure in LLM-integrated productivity tools by intercepting and analysing outbound traffic with Python and mitmproxy under controlled conditions.

Next.js | Supabase | PostgreSQL | TypeScript

- **Software & security engineering:** designed, built, and deployed a production full-stack booking platform with a security-first architecture — Postgres row-level security, strict Content Security Policy, MFA-gated admin, and audit logging — then conducted a self-audit that uncovered and remediated a privilege-escalation vulnerability.

MangoVision — Aerial Object Detection**Nov 2023 – Jul 2024**

PyTorch | YOLOv8 | Faster R-CNN | Detectron2

- **Deep learning:** developed and benchmarked YOLOv8 and Faster R-CNN detectors across a public benchmark and a custom drone dataset, reaching mAP@0.5 of 0.959 and 0.756. *Best Research Project Award, Gold Medal.*

PUBLICATIONS

M.H. Lye and **M.A. Eldeib**, “Performance Analysis of Faster R-CNN and YOLOv8 Model for Mango Fruits Detection,” *Journal of Informatics and Web Engineering*, vol. 5, no. 2, pp. 280–289, 2026. doi: doi.org/10.33093/jiwe.2026.5.2.17

TECHNICAL SKILLS

- **Security & Privacy:** Web Application Penetration Testing, Burp Suite, OWASP ZAP, Nmap, Wireshark, mitmproxy, OWASP Top 10, Network Traffic Analysis.
- **Security Monitoring (SIEM):** Splunk, IBM QRadar, log analysis, threat detection.
- **Programming:** Python, C/C++, Java, SQL, JavaScript, TypeScript, Flutter.
- **Backend & APIs:** Node.js, Spring Boot, REST APIs, GraphQL.
- **Web & Frontend:** React, Next.js, HTML, Tailwind CSS.
- **Databases:** PostgreSQL, Redis.
- **Cloud, DevOps & Tooling:** AWS (Lambda, API Gateway, DynamoDB, ELB, Auto Scaling), Docker, Git, GitHub Actions, Playwright, Linux (Kali, Ubuntu).
- **AI & Data Science:** PyTorch, scikit-learn, MCP.
- **Project Management:** Agile/Scrum, Jira.

Languages: English (C1) | German (B1).